

Dinuka Avindra Silva

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PROFESSIONAL SUMMARY

Passionate Software Engineering Intern with expertise in full-stack development, problem-solving, and building efficient, user-friendly applications. Specialised in secure system architectures, penetration testing concepts, and high-performance engineering. Driven by the challenge of bridging creative development with offensive security to build systems that are both functional and inherently resilient.

TECHNICAL SKILLS

Frontend: React.js, Next.js, JavaScript, TypeScript, Tailwind CSS, Material UI, Responsive Design

Backend: Java / Spring Boot, Node.js / Express, PHP / Laravel, FastAPI / Python

Databases: MySQL, PostgreSQL, MongoDB, Firebase, DynamoDB

Cloud & DevOps: AWS (Lambda, DynamoDB), Docker, Git / GitHub, Linux / Bash

Tools: Postman, Maven, Spring Security, REST API, WebSocket, JWT

AI / ML: OpenAI API, Scikit-learn, AWS AI Services, NLP

KEY PROJECTS

AI Chatbot System (LUMI) / AI / Node.js

Technologies: Node.js, Express, OpenAI API, React, REST API, WebSocket

- Built a full-stack chatbot application with a Node.js backend and React frontend following responsive design principles.
- Integrated OpenAI GPT API for advanced natural language understanding and context-aware responses.
- Implemented secure API key management and data encryption for user privacy protection.
- Designed real-time message streaming and conversation history management with efficient state handling.
- Created an intuitive UI with typing indicators, message timestamps, and smooth animations.

Smart Cart AI / Personal Full-Stack Project

Technologies: React, Tailwind CSS, Spring Boot, PostgreSQL, FastAPI, Python, Scikit-learn, JWT, Docker

- Developed a full-stack AI-powered e-commerce platform covering Electronics, Fashion, Home, Beauty, and Sports categories.
- Implemented secure JWT-based authentication and role-based access control for users and administrators.
- Designed and developed RESTful APIs with Spring Boot for product management, cart operations, order processing, and user management.
- Built an AI recommendation engine using Python, FastAPI, and Scikit-learn to provide personalised product suggestions based on user behaviour and purchase history.
- Implemented advanced features including Wishlist, smart search, product filtering, ratings & reviews, order tracking, and notifications.
- Created an admin analytics dashboard to monitor sales, revenue, top-selling products, and customer activity.
- Containerised application services with Docker to support scalable deployment and microservice architecture.

Multi-Vendor Marketplace Platform / Group Project (8-member team)

Technologies: React, Vite, Material UI, Framer Motion, Spring Boot, MongoDB, JWT

- Collaborated in an 8-member team to develop a scalable multi-vendor e-commerce marketplace.
- Implemented role-based modules for Customers, Sellers, and Administrators.
- Built backend REST APIs with Spring Boot and MongoDB for product, vendor, and order management.
- Participated in frontend enhancement, API integration, UI optimisation, and feature testing.
- Used GitHub for version control and collaborative development workflows.

Smart Library Management System / Java / Spring Boot

Technologies: Java, Spring Boot, MongoDB, REST API, Maven, Spring Security

- Engineered an enterprise-grade library management system with comprehensive resource tracking and role-based access control.
- Implemented a role-based authentication system for librarians, members, and administrators with granular permission controls.
- Built a book catalogue management system with advanced search, filtering, and categorisation capabilities.
- Created automated borrowing and return workflows with due date notifications and fine calculation systems.
- Developed RESTful APIs with proper exception handling and data validation for system integrity.

AI-Powered Study Planner / SaaS / Automation

Technologies: Next.js, React, TypeScript, Tailwind CSS, AWS Lambda, DynamoDB

- Developed an AI-powered scheduling algorithm that generates optimised study plans based on subject difficulty, deadlines, and learning pace.
- Integrated AWS cloud services for scalable backend infrastructure and AI model deployment.
- Created interactive progress tracking dashboards with visual analytics for academic performance monitoring.
- Built a notification system for deadline reminders and study session alerts.

CERTIFICATIONS

- **AWS Certified AI Practitioner** — Amazon Web Services (2026)
- **Generative AI Foundation** — Google Cloud (2026)
- **Python Programming** — University of Moratuwa (2023)

EDUCATION

BSc Information Technology

University of Vavuniya, Sri Lanka

REFERENCES

Mr. Jayasinghe

Retired AGM HR, Bank of Ceylon WPN
Phone: 077 353 9231

Dr. D. Kartheeswaran

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